

Supermarket Sales: Final Data Analysis Report

Combined review of the Excel, Python, SQL Server and Power BI work. All figures come from the project files; nothing was recalculated or re-analysed.

1. Executive Summary

This project analyses 1,000 supermarket transactions from three branches (Yangon, Mandalay, Naypyitaw) between 1 January and 30 March 2019. The business objective is to understand sales, profitability, customer behaviour and product returns, and to present the results in a Power BI dashboard.

307,391 Gross Sales	55,100 Gross Profit	17.9% Gross Profit Margin	1,000 Transactions	5,510 Units sold	25% Return rate
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- Sales are spread evenly across the six product lines (46,851 to 53,471). Food and beverages leads on sales; Electronic accessories leads on gross profit (9,710).
- One in four transactions (250 of 1,000) was returned. Food and beverages has the highest return rate at 30.5%; Home and lifestyle the lowest at 20.0%.
- February was the weakest month: 92,393 against 110,754 in January and 104,243 in March.
- Members and Normal customers generate similar sales (156,403 vs 150,987) and identical return rates (25%).
- Two points need attention before publishing: unit cost is derived as *Unit price – 10*, and the Average Transaction Value in Power BI is calculated on a different basis from the SQL query (see Section 8).

2. Project Overview

Project	Supermarket Sales analysis (file names: SuperMarket_Sales, SuperMarket_Sales_BI)
Business context	Retail sales with transaction-level data, plus a separate returns table
Records	1,000 sales rows (1,000 unique Invoice IDs) and 1,000 return records (yes / no)
Period	1 Jan 2019 to 30 Mar 2019 (Jan, Feb, Mar)
Main fields	Invoice ID, Branch, City, Customer type, Gender, Product line, Unit Cost, Unit price, Quantity, Tax 5%, Total, Date, Month, Time, Payment, Rating, Returns
Dimensions	3 branches (A Yangon, B Mandalay, C Naypyitaw), 6 product lines, 2 customer types, 2 genders, 3 payment methods. Ratings range 4.0 to 10.0.
Tools	Excel, Python (Pandas, Matplotlib, Seaborn), SQL Server (T-SQL), Power BI

Branch-to-city mapping is taken from the matching totals in the Excel pivots. The files do not state a currency, so values are shown without one.

3. Data Preparation

Only steps visible in the files are listed.

- **Excel:** the data sits in a formatted table (Table2) with a second sheet, Data2, holding the returns list. Calculated columns: Unit Cost (= Unit price – 10), Gross Sales (Unit price × Quantity), COGS (Unit Cost × Quantity), Gross Profit, Gross Profit Margin (with IFERROR), Month (from Date) and Return Flag (looked up from Data2; yes = 1, no = 0).
- **Python:** loaded both sheets with Pandas, checked structure with `info()` (1,000 rows, no missing values in any column), re-created Gross Sales, COGS, Gross Profit and Margin, and joined sales to returns on Invoice ID (left join). Months were ordered Jan, Feb, Mar.
- **SQL Server:** validation queries (preview, row counts, NULL checks on Invoice_ID, Product_Line, Unit_Cost, Unit_Price, Quantity).
- No duplicate removal, outlier treatment or type conversion beyond the above is documented.

4. Excel Analysis

The **PT** sheet contains 17 pivot tables and 2 doughnut charts. The **Dashboard** sheet holds 7 charts (3 bar, 1 line, 3 doughnut). Pivots cover Gross Sales, Gross Profit and Profit % by product line, monthly sales, branch by month, sales by city and by gender and customer type (Gross Sales and Total including 5% tax), transaction counts, quantity, average rating and return rate.

Product line	Gross Sales	Gross Profit	Profit %	Quantity	Avg rating	Return rate
Electronic accessories	51,750	9,710	18.8%	971	6.92	25.3%
Fashion accessories	51,720	9,020	17.4%	902	7.03	24.2%
Food and beverages	53,471	9,520	17.8%	952	7.11	30.5%
Health and beauty	46,851	8,540	18.2%	854	7.00	22.4%
Home and lifestyle	51,297	9,110	17.8%	911	6.84	20.0%
Sports and travel	52,301	9,200	17.6%	920	6.92	27.1%
Total	307,391	55,100	17.9%	5,510	6.97	25.0%

Cut	Results (Gross Sales)
Month	Jan 110,754 · Feb 92,393 · Mar 104,243
Branch / city	A Yangon 100,947 · B Mandalay 101,141 · C Naypyitaw 105,304
Customer type	Member 156,403 (501 transactions) · Normal 150,987 (499)

Cut	Results (Gross Sales)
Gender	Female 159,888 (501 transactions) · Male 147,502 (499)
Payment (transactions)	Cash 344 · Ewallet 345 · Credit card 311
Return rate by branch	A 23.5% · B 25.0% · C 26.5%

Excel also reports Total sales including 5% tax of 322,967.

5. Python Analysis

The notebook computes the headline KPIs and then groups results by product line, month, customer type, payment method, branch and city, with a Matplotlib or Seaborn chart for each. It also plots the rating distribution (histogram) and returns by product line.

Output	Value
Gross Sales	307,390.75
COGS	252,290.75
Gross Profit	55,100.00
Gross Profit Margin	17.93%
Quantity / Orders	5,510 / 1,000
Returned orders / Return rate	250 / 25.0%
Average rating	6.97

Other Python results: gross profit by branch A 18,590, C 18,310, B 18,200; gross sales by payment Cash 106,863, Ewallet 104,559, Credit card 95,969; returns by product line Food and beverages 53, Sports and travel 45, Fashion accessories 43, Electronic accessories 43, Health and beauty 34, Home and lifestyle 32. Every Python figure matches the Excel pivots.

6. SQL Analysis

The script (`supermarket_analysis.sql`, SQL Server syntax) works on two tables, **Sales** and **Returns**. It contains:

- **Validation:** row counts and NULL checks.
- **Overall KPIs:** Gross Sales, COGS, Gross Profit, Gross Profit Margin, Transactions, Average Transaction Value (Gross Sales ÷ distinct Invoice_ID).
- **By product line:** sales, COGS, profit, margin, quantity, transactions, average rating.
- **By segment:** customer type, gender, payment method, month (Jan to Mar), branch and city.
- **Returns:** overall return rate, and return rate by product line, customer type and gender, using a LEFT JOIN from Sales to Returns. A final query combines all product-line metrics in one summary.

Limitation: the SQL file contains queries only. No query outputs were saved, so SQL results could not be compared number by number with Excel, Python and Power BI. The metric definitions match the other tools, except Average Transaction Value (Section 8). The SQL return queries filter on Returns = 1, whereas Excel and Python store returns as "yes"/"no"; the SQL table was presumably converted, but this is not visible in the file.

Gender return rate is the only return breakdown that exists in the SQL script but not in the other tools.

7. Power BI Dashboard

The Power BI export has three pages. Page names are not visible in the PDF, so descriptive labels are used below.

Page 1: Executive overview

Purpose: headline performance at a glance. **KPIs:** Gross Sales 307K, Gross Profit 55K, Gross Profit % 18%, Transactions 1K, Avg Transaction Value 323, Return Rate 25%. **Visuals:** Gross Sales by month, Gross Sales by product line, Return Rate by product line, and three callouts (Top Sales: Food and beverages; Top Profit: Electronic accessories; Lowest Sales Month: February). **Slicers:** Date, Customer Type, Product Line, Payment. **Insight:** a Key Findings box flags Food and beverages return rate of 30% as a risk, 5 points above overall.

Page 2: Product line performance

Purpose: compare profitability by product line. **KPIs:** Gross Sales 307K, Gross Profit 55K, Gross Profit % 18%, COGS 252K. **Visuals:** a table (sales, profit, profit %, transactions, avg transaction value, return rate) and bar charts for Gross Sales, Gross Profit, COGS and Gross Profit % by product line. No slicers are visible. **Insight:** margins are close (17% to 19%); Electronic accessories is highest at 19%, Fashion accessories lowest at 17%.

Page 3: Customer and returns

Purpose: understand who buys, how they pay and where returns occur. **Visuals:** customer-type table, Gross Sales and Gross Profit by gender, Return Rate by customer type, Returned Transactions and Return Rate by product line, Gross Sales by payment. No slicers are visible. **Insight:** Members and Normal customers both return 25%; Food and beverages has the most returned transactions (53).

8. Key Findings

Finding: Food and beverages has the highest return rate

Evidence: 30.5% (53 of 174 transactions) vs 25.0% overall; Home and lifestyle is lowest at 20.0%.

Business meaning: the top-selling category also has the most returns, so part of its sales does not stay as revenue. The files do not show why.

Finding: one in four transactions is returned

Evidence: 250 of 1,000 (25.0%). By branch: A 23.5%, B 25.0%, C 26.5%.

Business meaning: returns are a material issue across the business, not limited to one segment.

Finding: February was the weakest month

Evidence: Gross Sales Jan 110,754, Feb 92,393 (–16.6% vs Jan), Mar 104,243. February also had 303 transactions vs 352 in January and 345 in March.

Business meaning: the dip comes from fewer transactions. With only three months of data, seasonality cannot be concluded.

Finding: product lines perform closely, with Health and beauty trailing

Evidence: Gross Sales range 46,851 (Health and beauty) to 53,471 (Food and beverages); Health and beauty also has the fewest transactions (152).

Business meaning: no single category dominates; Health and beauty is the smallest contributor.

Finding: Electronic accessories earns the most profit

Evidence: Gross Profit 9,710 and margin 18.8%, vs Food and beverages 9,520 with the highest sales.

Business meaning: the highest-selling and highest-profit categories differ, but only narrowly.

Finding: Members and Normal customers behave alike

Evidence: Gross Sales 156,403 vs 150,987 (+3.6%); 501 vs 499 transactions; return rate 25% for both; margin 18% for both.

Business meaning: membership is associated with slightly larger baskets but not with fewer returns.

Finding: Naypyitaw leads sales, Yangon leads profit

Evidence: Sales C Naypyitaw 105,304 vs 100,947 to 101,141 elsewhere; profit A Yangon 18,590 vs 18,310 (C) and 18,200 (B).

Business meaning: branch differences are small (about 4%).

Finding: credit card payments are the smallest channel

Evidence: Gross Sales 95,969 and 311 transactions vs Cash 106,863 (344) and Ewallet 104,559 (345).

Business meaning: cash and e-wallet are used more often than credit card.

Data points to resolve

1. **Average Transaction Value differs by definition.** Power BI shows 323, which equals Total including 5% tax ÷ 1,000 (322,967 ÷ 1,000). The SQL query divides Gross Sales by transactions, which gives 307.39 (307,391 ÷ 1,000). The same dashboard page shows Gross Sales of 307K, so 323 looks inconsistent to a reader. Pick one definition and label it.

2. **Unit Cost is derived, not sourced.** In Excel, Unit Cost = Unit price – 10, and this holds in all 1,000 rows. Gross Profit is therefore exactly 10 per unit (5,510 units × 10 = 55,100). Margin differences between product lines reflect price levels only. If real cost data exists, profit and margin results would change.

3. **Minor labels:** "Groos Profit" (Excel column), "Cutomer Type" (Power BI slicer) and "Dashbord" (Excel sheet) are misspelled.
- Excel, Python and Power BI otherwise agree; differences are rounding only (for example 18% vs 17.93%).

9. KPI Summary

KPI	Value	Description	Source
Gross Sales	307,390.75	Unit price × Quantity, before 5% tax	Excel, Python, Power BI
COGS	252,290.75	Unit Cost × Quantity	Excel, Python, Power BI
Gross Profit	55,100	Gross Sales – COGS	All tools
Gross Profit Margin	17.93%	Gross Profit ÷ Gross Sales	All tools
Total incl. 5% tax	322,966.75	Sum of the Total column	Excel
Quantity Sold	5,510	Total units	Excel, Python
Transactions	1,000	Distinct Invoice IDs	All tools
Avg Transaction Value	323 (PBI) / 307.39 (SQL basis)	See discrepancy 1	Power BI, SQL
Returned Transactions	250	Invoices marked "yes"	Excel, Python, Power BI
Return Rate	25.0%	Returned ÷ total transactions	All tools
Average Rating	6.97	Mean customer rating	Excel, Python

Discount and Net Sales do not exist in this project and are not reported.

10. Business Recommendations

- **Investigate Food and beverages returns first.** It has 53 returned transactions and a 30.5% rate. Return reasons are not in the dataset, so collecting them (for example, damaged, expired or wrong item) is needed before choosing a fix.
- **Review Sports and travel (27.1%) alongside it, and compare with Home and lifestyle (20.0%)** to see whether practices differ. This requires data outside the current files.
- **Check the February drop.** It coincides with 303 transactions vs about 350 in the other months. Add more months before treating it as seasonal.
- **Review Health and beauty.** It has the lowest sales and fewest transactions; foot traffic, range and pricing data are needed to explain this.
- **Look at credit card usage.** It trails cash and e-wallet by roughly 33 transactions each. Whether this is a checkout issue or customer preference cannot be told from this data.
- **Validate cost data.** Replace the derived Unit Cost with actual purchase costs to make profit and margin recommendations reliable.
- **Standardise Average Transaction Value** across Power BI and SQL, and correct the spelling errors before sharing the dashboard.

11. Conclusion

The project analysed 1,000 supermarket transactions across three branches and six product lines over Q1 2019. Sales of 307,391 produced a gross profit of 55,100 (17.9%). Performance is balanced across categories, branches and customer types. The clearest issue is returns: 25% of transactions, rising to 30.5% in Food and beverages. February sales were also weaker. The business can learn where to focus its attention, but explaining the causes needs return reasons, actual cost data and a longer time period. The four tools produce consistent results, apart from the Average Transaction Value definition.

12. Tools & Skills

Area	Skills shown in the files
Excel	Pivot tables (17), calculated columns, XLOOKUP, IFERROR, charts, Excel dashboard sheet
Data preparation	Data validation, derived fields, joining tables, exploratory data analysis
Python	Pandas (groupby, merge, info), Matplotlib, Seaborn
SQL Server	T-SQL queries, aggregations, CASE, LEFT JOIN, CAST, NULLIF, data validation
Power BI	Three-page dashboard, slicers, KPI cards, table and bar visuals, visual design
Analytics	KPI analysis, data visualization, business insights and recommendations

NumPy is imported in the notebook but not used in the visible code, so it is not listed as a demonstrated skill.

Project Completion Summary

- ✓ Excel: workbook with 17 pivots and a dashboard sheet
- ✓ Python: notebook with KPIs and charts

- ✓ SQL: full query script (outputs not included)
- ✓ Power BI: 3-page dashboard (PDF export; the .pbix file was not uploaded)
- ✓ Business Insights: findings and recommendations above
- ✓ Final Report: this document